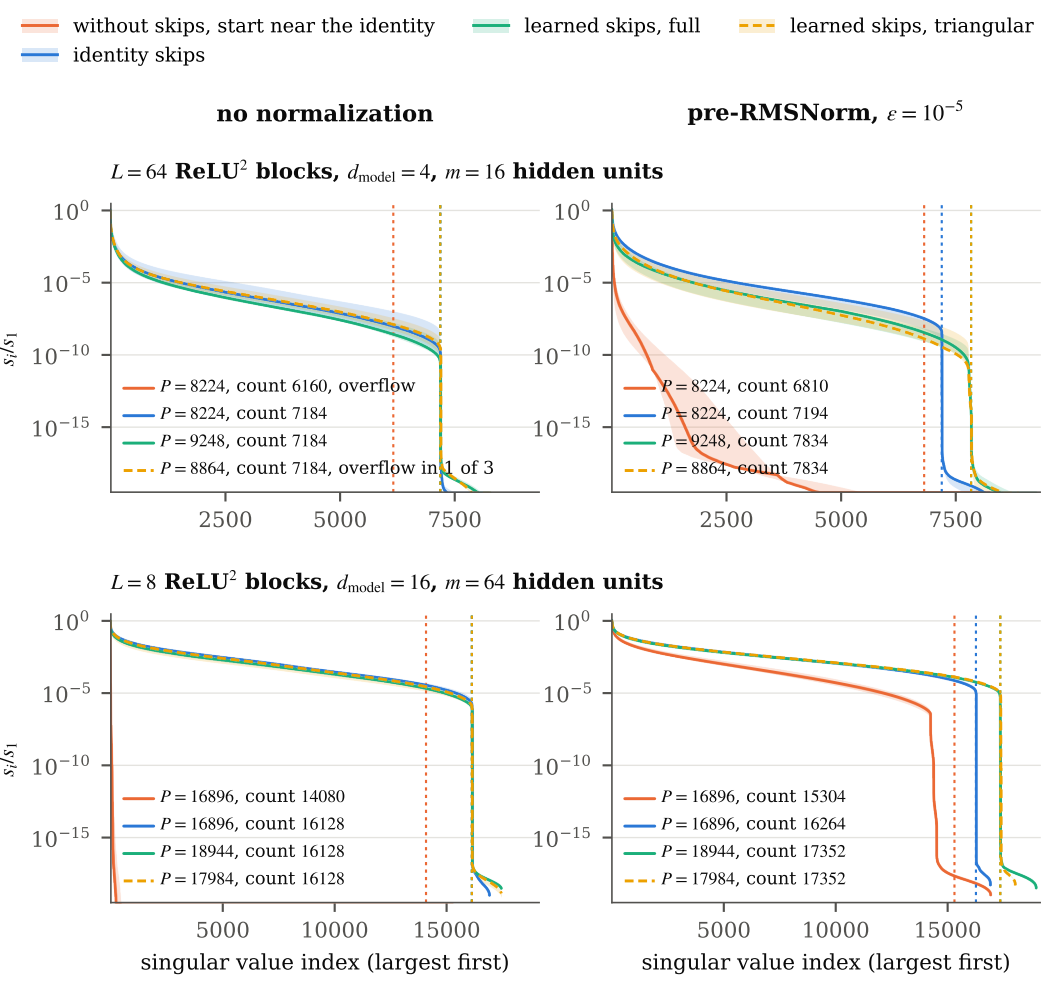

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ ReLU^2 blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 8224$, count 6160, overflow
$P = 8224$, count 7184
$P = 9248$, count 7184
$P = 8864$, count 7184, overflow in 1 of 3
$P = 8224$, count 6810
$P = 8224$, count 7194
$P = 9248$, count 7834
$P = 8864$, count 7834
2500
5000
7500
$L = 8$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$P = 16896$, count 14080
$P = 16896$, count 16128
$P = 18944$, count 16128
$P = 17984$, count 16128
$P = 16896$, count 15304
$P = 16896$, count 16264
$P = 18944$, count 17352
$P = 17984$, count 17352
5000
10000
15000
singular value index (largest first)
singular value index (largest first)